# TECHNOTRONICS™

***Innovating the Future***

***See Beyond the Fog — Drive Safer, Day or Night with Driversense***

## Every Second Counts When Visibility Fails

# Overview

Poor visibility causes thousands of severe road accidents every year. Fog, rain, dust, and nighttime driving drastically reduce reaction time—putting drivers, fleets, and families at risk.

The **Technotronics Advanced Laser Detection System** provides long-range, ultra-precise obstacle detection using high-intensity 905 nm laser technology. Unlike cameras or radar, the system performs reliably even in **dense fog**, **low-light conditions**, and **complex road environments**.

This technology gives drivers and fleet operators the additional seconds they need to prevent collisions and save lives.

# The Fatal Gap: Why Human Sight Fails

Modern infrastructure has outpaced human biology. When visibility drops, our natural senses cannot bridge the gap.

## The Science of Visibility Failure

- **Contrast Collapse:** Fog erases depth; by the time you see an object, you've already run out of braking distance.
- **Photonic Scattering:** Rain and snow act as mirrors, reflecting light back to mask road hazards and lane markings.
- **Depth Deficit:** Low light triggers the "Purkinje effect," causing the brain to misjudge an obstacle's speed and distance

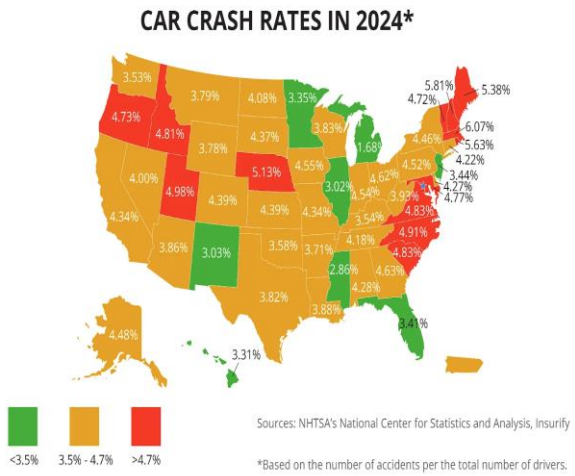
  **The Chain Reaction:** Multi-vehicle pileups are a failure of physics, not intent. At high speeds, the "reaction window" vanishes—drivers often collide before they can even touch the brake pedal.

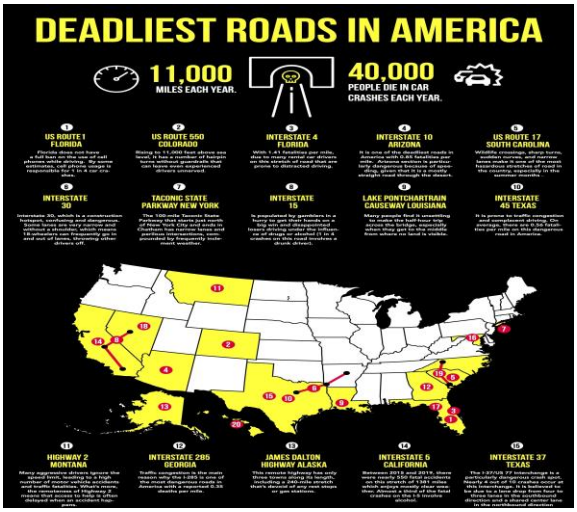
# Regional Analysis: Fog & Collision Hotspots

## United States: High-Speed Vulnerability

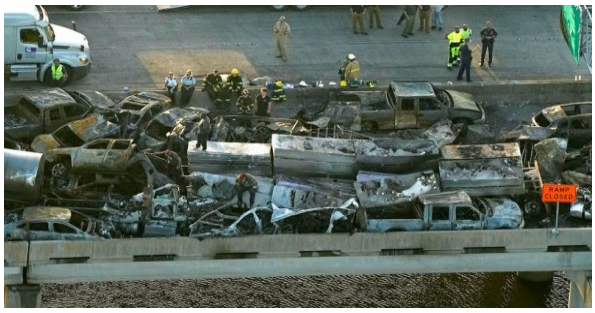
In the U.S., the combination of **driver fatigue** and sudden **valley or coastal fog** contributes to over 1.2 million weather-related crashes annually.

- **Key Hotspots:** The Gulf Coast, Pacific Northwest, and the Appalachian Valley.
- **The Impact:** Multi-vehicle "chain-reaction" pileups on high-speed interstates, leading to massive insurance write-offs and long-term litigation.



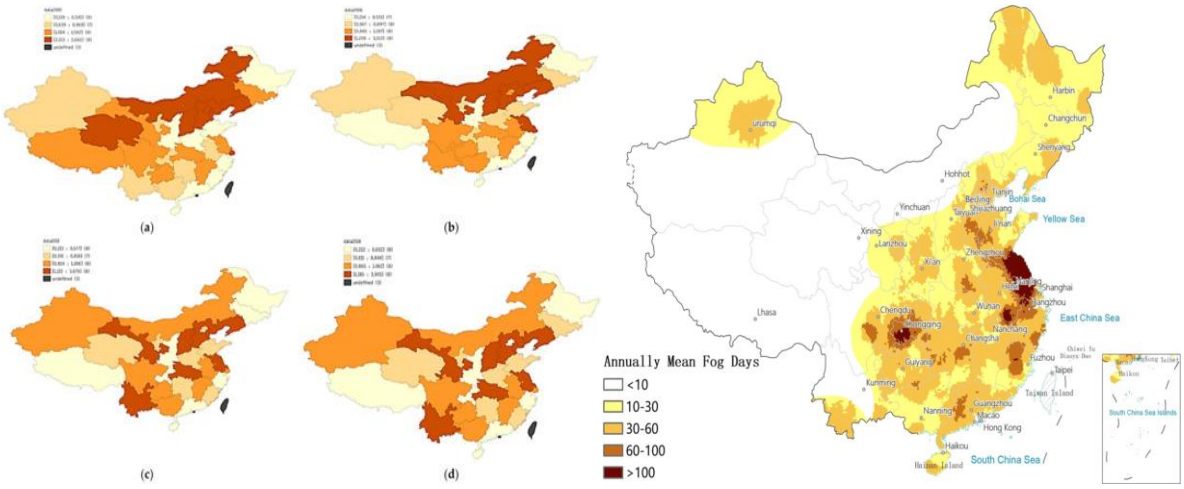

## Highway vehicles pileup



## China: Rapid Growth & Coastal Hazards

China faces unique challenges where high-density industrial corridors meet heavy seasonal fog.

- **Key Hotspots:** Eastern and Southern provinces (e.g., Jiangsu, Zhejiang, and Shandong).
- **The Impact:** Historically, these regions have seen some of the world's largest multi-vehicle collisions (often involving 50+ vehicles) due to the sheer volume of commercial trucking and sudden visibility drops to near zero.



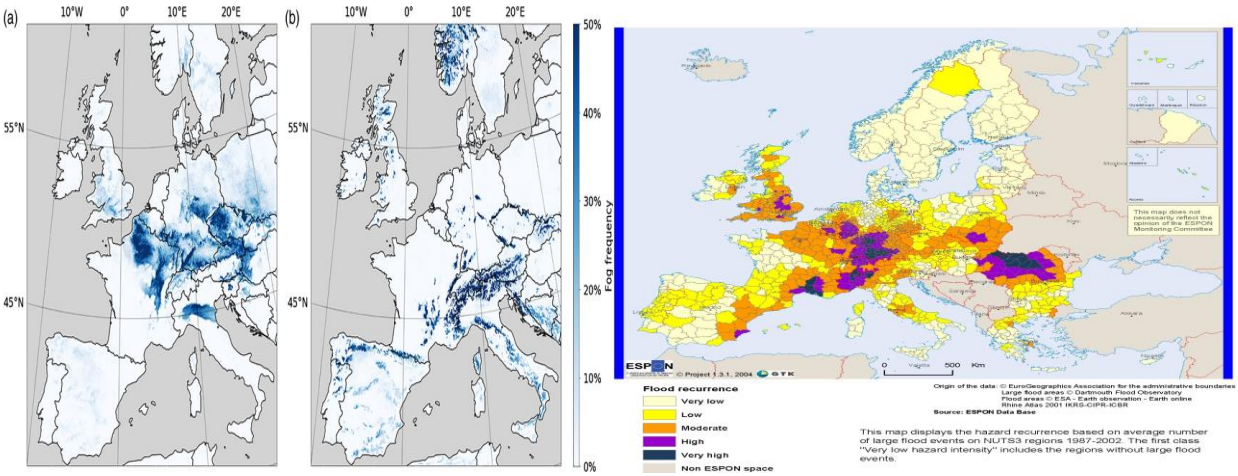

## Highway Pileups

Showing regions in eastern and southern China prone to heavy fog that has historically caused multi-vehicle collisions.

## Europe: The Tech-Safety Nexus

European corridors often suffer from "patchy" fog where visibility changes within meters, catching even experienced drivers off guard.

- **Key Hotspots:** The Po Valley (Italy), Benelux regions, and the German Autobahn.
- **The Solution:** This region serves as a prime case study for why **roadside sensors** and **V2X (Vehicle-to-Everything)** visibility warnings are critical for reducing "stop-and-go" waves.




## Highway Pileups

European case studies where fog and driver fatigue combine — great illustration of why visibility warnings and road sensors matter.

# The Next Generation of Safety Vision

Our laser-based detection system measures distance and identifies obstacles from 400, 1000 up to 1500 meters — even in fog, rain, or total darkness. It helps drivers react earlier, avoid collisions, and stay safe when cameras, radar, and human eyes fail.

Our Driversense V.1 is a laser system that penetrates these visual obstructions and reveals what the eye cannot see. It conveys the real time data to the receiver that displays obstacles distance, speed, collision time, and other environmental details like temperature and Humidity.

## Core Capabilities

### ✓ 400m Long-Range Obstacle Detection

Developed for unpredictable conditions—fog, rain, night, low contrast.

### ✓ High-Speed Laser Pulse Processing

Fast response output helps reduce driver reaction time.

### ✓ Fog-Penetration Technology

Engineered for clarity in low-visibility environments.

### ✓ Real-Time Distance Measurement

Units transmit live data to dashboards or vehicle systems.

### ✓ Scalable Detection Models

Our detection family includes future 400 m, 1000 m, and 1500 m variants for expanded applications.

### ✓ Optical Enhancements (New)

- **Collimating lenses** for tighter beam divergence and longer usable range
- **Polycarbonate protective films** to shield optics from debris, scratches, and UV damage

# Engineered for Any Distance – From City Roads to Open Highways

Our upcoming models extend detection capability from 400m, 1000 m, 1500 m, and up to 2000 m — ensuring superior safety in any environment. These modular versions fit every use case, from standard vehicles to industrial fleets and autonomous systems.

| Feature | Module 1 (TC25-400) | Module 2 (TC25-1000) | Module 3 (TC22-1500) |
|---|---|---|---|
| **Module Image** | | | |
| **Measuring Range** | 3 - 400m | 3 - 1000m | 3 - 1500m |
| **Accuracy** | ±1m | ±1m | ±1m |
| **Frequency** | 1 - 3Hz | 1 - 3Hz | Up to 100Hz |
| **Resolution** | 0.1m | 0.1m | 0.1m |
| **Laser Type** | 905nm (Pulse) | 905nm (Pulse) | 905nm (Pulse) |
| **Voltage** | DC +3.3V | DC +3.3V | DC +3.3V |
| **Communication** | TTL/RS232/RS485/BT | TTL/RS232/RS485/BT | TTL/RS232/RS485/BT |
| **Dimensions** | 43 x 22mm | 43 x 22mm | 43 x 22mm |
| **Weight** | 15g | 15g | 15g |

# For Every Road, Every Driver

• Fleet logistics and long-haul transport • Passenger cars and dealerships • Emergency cars and rescue vehicles • Mountain and fog-prone areas.

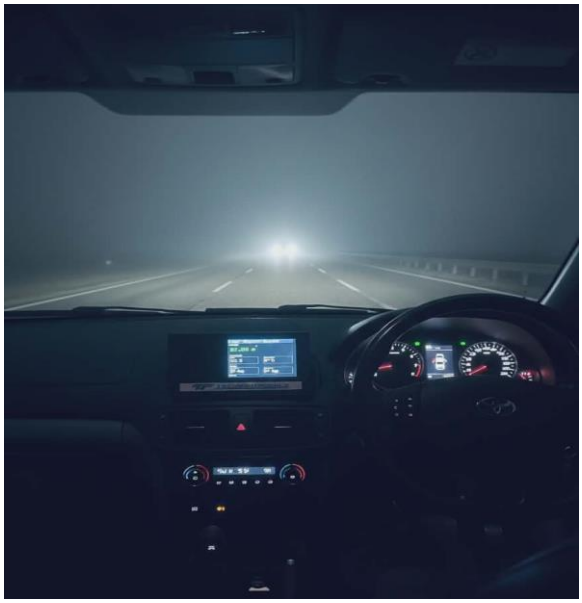
## A Modular Road-Safety Solution

From compact vehicles to heavy-duty transport fleets, each system can be adapted with optional accessories such as strengthened enclosures, collimating optics, protective films, and extended-range modules.

## Designed for Every Environment

**Applications:**

- Long-haul freight and logistics fleets
- Passenger vehicles / dealerships
- Mountain regions and fog-prone locations
- Emergency and rescue vehicles
- Industrial transport & autonomous platforms.



## One Device Can Save a Life

Whether for professional fleets or family cars, our mission is the same — to make roads safer for everyone, everywhere.

Its advanced laser system detects obstacles in real-time and alerts the driver even in adverse environmental conditions like fog, or low visibility.

# Contact Information:

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